

Device Introduction

Breakthroughs in AR Computing Platform Technologies



X3 Pro

- ♦ The first 76g lightweight binocular full-color AR smart glasses
- ♦ Equipped with the ultra-compact Optical Engine + photolithographic etched waveguide
- ♦ World-leading intent recognition powered by AI
- ♦ Snapdragon AR1 chip, purpose-built for AR glasses



X2

- ♦ World's first true full-color consumer-grade AR glasses
- ♦ First to feature Micro-LED + diffractive waveguide full-color display
- ♦ 16.7 million colors with binocular independent full-color output
- ♦ 1500 nits peak to-eye brightness



X1 Pioneer Ver.

- ♦ Holographic single-layer waveguide structure
- ♦ Proprietary wave coupling analysis and optimization algorithms
- ♦ Pioneered full-color microdisplay engine
- ♦ Supports notifications, photo sharing, photo translation, and more

RayNeo X Series AR glasses provide developers with a hardware platform featuring full-color display, high brightness, and extreme lightness. Combined with high-performance computing and open interfaces, they empower efficient development and deployment of innovative AR applications.

Glasses Overview

RayNeo X3 Pro

RayNeo X3 Pro is the flagship AR glasses released by RayNeo in 2025. Based on the new Qualcomm Snapdragon AR1 platform, it adopts the world's smallest mass-produced full-color MicroLED optical engine, significantly enhancing display brightness and color performance. With lighter weight, upgraded interaction methods, and greatly enhanced AI capabilities and application ecology, it targets high-end consumer and professional markets.



RayNeo X2

RayNeo X2 is a consumer-grade true AR glasses innovatively launched by RayNeo, featuring a lightweight integrated design. Equipped with the Qualcomm Snapdragon XR2 platform and a MicroLED +光波导 (diffractive waveguide) display solution, it supports multiple interaction methods (temple touch, smart ring, voice), making it suitable for daily wear and in-depth developer experience.

Core Specifications Comparison

Specification/Model	RayNeo X3 Pro	RayNeo X2
Chip Platform	Qualcomm Snapdragon AR1 Gen1	Qualcomm Snapdragon XR2
Storage Configuration	4GB RAM + 32GB ROM	6GB RAM + 128GB ROM
Display Solution	0.36cc full-color MicroLED optical engine + diffractive waveguide	Diffractive waveguide + full-color MicroLED
Refresh Rate	Up to 60Hz	Up to 60Hz
Resolution	640×480	640×480
FOV (Field of View)	30°	25°
PPD (Pixels Per Degree)	27	32

Brightness	Peak 6000nits, Average 3500nits	Peak >1500nits
Camera	12MP main camera (RGB) + spatial camera (VGA)	Single camera: 16MP, 1080p video
Sensors	- IMU sensors: gyroscope, accelerometer, magnetometer - Removal sensor: one on left temple	- IMU sensors: gyroscope, accelerometer, magnetometer, GPS - Removal sensor: one on right temple
Speakers	2 built-in stereo speakers, support reverse noise cancellation, stereo output	Dual speakers on left/right temples
Microphones	3 microphones (one on each temple, one in front)	3 microphones, support directional sound pickup
Interaction Methods	Temple touch, voice, watch/phone interaction	Temple touch, smart ring, voice
Weight	76g	119g
Dimensions	L153.16mm × W45.65mm × H168.6mm	--
Connectivity	- Wi-Fi 6: supports 2.4G&5G dual-band, 802.11 a/b/g/n/ac - Bluetooth 5.2	- Wi-Fi 5 - Bluetooth 5.2
USB	USB 2.0, Type C interface	--
Battery/Lifespan	245mAh, 3-5 hours moderate use	590mAh, 2-3 hours moderate use
Charging	5V voltage, max charging current 735mA	5V voltage
Myopia Solution	- Snap-on lenses - Full-fit lenses	Snap-on lenses
Operating System	RayNeo AI OS 2.0	RayNeo OS

 [Return to the homepage](#)